

Problem Statement

A small college library maintains records of books manually. Your task is to create a JavaScript program that stores book information and generates useful reports for the librarian.

The program should use an array of objects to store book details and perform different operations using loops and conditional statements.

Book Information

Store details of at least **15 books**.

Each book should contain:

```
{  
  bookId: 101,  
  title: "B Programming",  
  author: "Dennis Ritchie",  
  category: "Programming",  
  copies: 5  
}
```

Task 1: Display All Books

Display complete information of every book available in the library.

Expected Output Format

Book ID: 101
Title: Introduction to Programming
Author: A. Sharma
Category: Programming
Copies Available: 5

Task 2: Find Total Number of Books

Calculate and display the total number of book records available in the library.

Example

Total Books: 15

Task 3: Books with Low Availability

Display books having fewer than 3 copies available.

Example

Database Management - 2 Copies
Computer Networks - 1 Copy

Task 4: Search Book by Book ID

Accept a Book ID and display the corresponding book details.

Example

Enter Book ID: 105

Book Found
Title: Data Structures
Author: Shaums

If the book does not exist, display:

Book Not Found

Task 5: Category-wise Book Count

Display the number of books available in each category.

Example

Programming: 4
Database : 3
Networking: 2
Mathematics : 3

Task 6: Book with Maximum Copies

Find and display the book having the highest number of copies.

Example

Most Available Book:

Task 7: Book Availability Status

Display status according to available copies.

Copies	Status
5 or more	Available
3 to 4	Limited
Less than 3	Critical

Example

Java Basics - Available
Data Structures - Limited
Computer Networks - Critical

Task 8: Count Books in Each Status

Display the number of books falling under each status.

Example

Available : 5
Limited : 4
Critical : 3

Task 9: Display Programming Books

Display all books belonging to the Programming category.

Task 10: Generate Library Summary Report

Display a final summary in the following format:

----- LIBRARY REPORT -----

Total Book Records : 12

Available Books : 5

Limited Books : 4

Critical Books : 3

Highest Copies Available : 10

----- END OF REPORT -----